

ssv-prom-exporter — Docker Compose guide

A Prometheus exporter for **DataCore SANsymphony** (SSV) that scrapes the SSV REST API and exposes inventory, health and performance metrics on `/metrics`. This guide is **Docker-Compose-first**: the fastest path to a working SANsymphony monitoring stack — Prometheus, Grafana, five ready-made dashboards, and optionally the exporter itself and the `prom-clip` companion — is the bundled `deploy/` compose project.

Who this is for. Operators who want SANsymphony observability without hand-wiring Prometheus and Grafana. Native Windows-service and Linux-systemd installs are covered briefly in Running without compose for production nodes that don't run a container runtime.

The exporter runs anywhere with TCP/443 reachability to a SSV management server — it does **not** have to run on the SSV host itself. All screenshots in this guide were captured live against a DataCore PSP 20 lab through this exact compose stack.

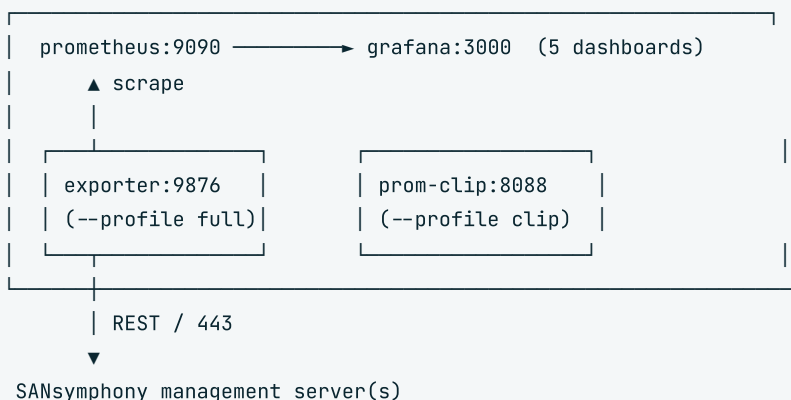
The compose stack at a glance

Everything lives under `deploy/`. The stack is built around **three optional services selected by compose profiles**, so you only run what you need:

Service	Profile	Host port	Role
prometheus	<i>always</i>	9090	Scrapes the exporter(s); 15 d retention.
grafana	<i>always</i>	3000	Five SSV dashboards, anonymous Viewer on.
exporter	<i>full</i>	<i>internal</i>	Runs <code>ssv-prom-exporter</code> itself against one SSV group.
prom-clip	<i>clip</i>	127.0.0.1:8088	Clip a Prometheus time window and replay it elsewhere.

Profiles compose freely — `--profile full --profile clip` brings up all four. With no profile flag you get just Prometheus + Grafana, which is Scenario A.

deploy/docker-compose.yml



Requirements

- A Docker host with **Compose v2** (`docker compose` , not the legacy `docker-compose`).
- DataCore SANsymphony **PSP 20+** (older may work, untested).
- TCP/443 reachability from wherever the exporter runs to the SSV management server.
- Outbound network from the Docker host to your exporter targets (Scenario A) or to the SSV REST API (Scenario B).

Quick start

You already run one or more exporters and just want Prometheus + Grafana:

```
# clone and enter the compose project
git clone https://github.com/lblanc/ssv-prom-exporter.git
cd ssv-prom-exporter/deploy
cp .env.example .env
# point EXPORTER_TARGETS at your exporter(s), e.g.
#   EXPORTER_TARGETS=lab=10.12.110.11:9876,prod=10.0.0.5:9876
$EDITOR .env
docker compose up -d
```

Grafana → `http://localhost:3000` (anonymous read-only, no login). Prometheus → `http://localhost:9090` .

You don't run an exporter yet and want the stack to host it too:

```
cd ssv-prom-exporter/deploy
cp .env.example .env
# set SSV_URL / SSV_USER / SSV_PASS / SSV_GROUP in .env
$EDITOR .env
docker compose --profile full up -d --build
```

The next three sections expand each scenario with full `.env` and command examples.

Scenario A — monitor exporters that already run elsewhere

default profile

The exporter runs on each SAN host (as a Windows service / systemd unit / standalone container); the compose stack only adds Prometheus + Grafana and scrapes those exporters over the network. This is the normal production topology.

1. Configure `deploy/.env`

```
# One entry per SANSymphony group: "groupname=host:port,...".
# groupname becomes the Prometheus `group` label the dashboards filter on.
EXPORTER_TARGETS=LAB-PVE=10.12.110.11:9876,HCI104=10.12.104.121:9876

# Grafana admin password (anonymous Viewer is always on; this gates editing).
GF_ADMIN_PASSWORD=admin
```

A single target is just as valid: `EXPORTER_TARGETS=lab=10.12.110.11:9876`.

2. Bring it up

```
cd deploy
docker compose up -d
docker compose ps
```

3. Verify every target is scraped

```
curl -s http://localhost:9090/api/v1/targets \
  | jq -r '.data.activeTargets[] | "\(.labels.group)\t\(.scrapeUrl)\t\(.health)'"
```

Or open Prometheus → **Status** → **Target health**. Each exporter target carries its `group` label and should read **UP**:

Prometheus Target health interface showing the status of targets.

prometheus 1 / 1 up

Endpoint	Labels	Last scrape	State
http://localhost:9090/metrics	instance="localhost:9090" job="prometheus"	333ms ago 8ms	UP

ssv-prom-exporter 2 / 2 up

Endpoint	Labels	Last scrape	State
http://10.12.110.11:9876/metrics	group="LAB-PVE" instance="10.12.110.11:9876" job="ssv-prom-exporter"	2.176s ago 849ms	UP
http://10.12.104.121:9876/metrics	group="HCI104" instance="10.12.104.121:9876" job="ssv-prom-exporter"	4.662s ago 1.361s	UP

Prometheus **Target health** — the **ssv-prom-exporter** job shows **2/2 up**, one target per SANsymphony group (**group="LAB-PVE"** , **group="HCI104"**).

4. Open Grafana

Browse to <http://localhost:3000> → **Dashboards** → **SSV**. Five dashboards are pre-provisioned; pick a group from the **Group** dropdown on any of them.

Grafana SSV folder view showing pre-provisioned dashboards.

SSV
Manage folder dashboards and permissions

Search for dashboards and folders

Filter by tag: ☐ Starred

Name	Tags
SSV — Hosts	datacore hosts ssv
SSV — Overview	datacore overview ssv
SSV — Ports	datacore ports ssv
SSV — Servers	datacore servers ssv
SSV — Storage	datacore ssv storage

The provisioned **SSV** folder: Overview, Servers, Storage, Hosts and Ports.

Scenario B — run everything in the stack

--profile full

For demos, PoCs, or sites that prefer to run everything containerized: the `full` profile adds the `exporter` service to the stack, so you install nothing on the SAN host. Prometheus auto-discovers it on the compose network (`exporter:9876`).

1. Configure `deploy/.env`

```
# --- exporter against one SSV group ---
SSV_URL=https://10.12.110.11      # SSV REST base URL (the mgmt server)
SSV_USER=administrator
SSV_PASS=ChangeMe!
SSV_GROUP=lab                    # becomes the Prometheus `group` label

# Optional failover knobs (see "HA & multi-group")
# SSV_BASES=10.12.110.12,10.12.110.13
# SSV_BACKUP_CIDRS=10.12.110.0/24

# Leave EXPORTER_TARGETS unset — Prometheus auto-targets exporter:9876.
GF_ADMIN_PASSWORD=admin
```

2. Bring it up

`--build` builds the exporter image locally from the repo. Drop it to pull `ghcr.io/lblanc/ssv-prom-exporter:latest` instead.

```
cd deploy
docker compose --profile full up -d --build
```

3. Watch it come alive

```
docker compose logs -f exporter
docker compose exec exporter wget -qO- http://127.0.0.1:9876/metrics | grep '^ssv_up'
# ssv_up{collector="health"} 1
# ssv_up{collector="inventory"} 1
# ssv_up{collector="performance"} 1
```

Exposing `/metrics` on the host. It is internal by default. To curl it from the host while in `--profile full`, uncomment the `ports`: block on the `exporter` service in `deploy/docker-compose.yml` (- `"9876:9876"`).

Pin a released image instead of building

```
# in deploy/.env  
EXPORTER_IMAGE=ghcr.io/lblanc/ssv-prom-exporter:v0.8.1
```

```
docker compose --profile full up -d      # no --build → pulls EXPORTER_IMAGE
```

Scenario C — add the prom-clip companion

--profile clip

`prom-clip` clips a time window out of one Prometheus and replays it into another (gzipped OpenMetrics out, remote-write in). The `clip` profile runs its web UI alongside the stack, bound on host loopback `127.0.0.1:8088` — no Windows Firewall prompt, no external exposure.

```
cd deploy
docker compose --profile clip up -d --build
# UI: http://127.0.0.1:8088
```

Combine with `--profile full` to run the exporter, Prometheus, Grafana **and** prom-clip together:

```
docker compose --profile full --profile clip up -d --build
```

prom-clip [Connection](#) [Export](#) [Import](#) [S3 target](#) [Status](#)

Prometheus connection

A single endpoint is used by both modes: as the source in [Export](#) and as the remote-write target in [Import](#).

URL

Username (optional)

Password (optional)

☐ Skip TLS verification

[Save](#) [Test connection](#)

Connection — a single Prometheus endpoint is the source for Export and the remote-write target for Import. From the compose network use `http://prometheus:9090` ; for a Prometheus on the docker host use `http://host.docker.internal:9090` .

To **import** a clip back into the stack's Prometheus, it must accept remote-write. That is an option, set in `deploy/.env` :

```
PROM_REMOTE_WRITE=1    # turns on --web.enable-remote-write-receiver
PROM_000_WINDOW=7d     # out-of-order window; old samples land instead of being dropped
```

```
docker compose up -d    # re-create prometheus with the receiver enabled
```

Without `PROM_000_WINDOW`, any sample older than the ~2 h head block is discarded while the write still returns `200 OK` — the import looks successful but lands nothing. See `prom-clip` in depth.

.env reference

All knobs read by `deploy/docker-compose.yml`. Copy `deploy/.env.example` to `deploy/.env` and edit. Everything is optional except the per-scenario essentials.

Variable	Profile	Default	Description
EXPORTER_TARGETS	default	lab=host.docker.internal:9876	Comma list <code>name=host:port</code> . Each <code>name</code> becomes the <code>group</code> label.
SSV_URL	full	<i>required</i>	SSV REST base URL, e.g. <code>https://10.0.0.1</code> .
SSV_USER	full	<i>required</i>	SSV username.
SSV_PASS	full	<i>required</i>	SSV password.
SSV_GROUP	full	<code>full</code>	Label applied to the in-stack exporter's metrics.
SSV_BASES	full	<i>empty</i>	Cold-start backup IPs (comma list) seeded before the first scrape.
SSV_BACKUP_CIDRS	full	primary's /24	CIDR allowlist for discovered backups. <code>0.0.0.0/0</code> disables filtering.
EXPORTER_IMAGE	full	<code>ghcr.io/lblanc/ssv-prom-exporter:latest</code>	Image to run when not using <code>--build</code> .
GF_ADMIN_PASSWORD	always	<code>admin</code>	Grafana admin password (editing only; viewing is anonymous).
PROM_REMOTE_WRITE	always	<i>off</i>	Set to <code>1</code> to accept inbound remote-write (backfill from prom-clip).
PROM_OOO_WINDOW	always	<code>7d</code>	Out-of-order storage window; required for backfill to land.
PROM_CLIP_IMAGE	clip	<code>ghcr.io/lblanc/prom-clip:latest</code>	Image for the prom-clip service when not using <code>--build</code> .

Exporter settings (env vars & flags)

Inside the stack (`--profile full`) the exporter is configured through the `SSV_*` env vars above. Run standalone (docker run, systemd, Windows service) the same settings are available as env vars and/or CLI flags, and can also come from a YAML config.

Flag	Env var	Description
<code>-config</code>	<code>SSV_CONFIG</code>	Path to a YAML config file (<code>config.example.yaml</code> has the full schema).
<code>-url</code>	<code>SSV_URL</code>	SSV REST base URL.
<code>-user</code>	<code>SSV_USER</code>	SSV username.
<code>-pass</code>	<code>SSV_PASS</code>	SSV password.
<code>-host</code>	<code>SSV_HOST</code>	<code>ServerHost</code> header; defaults to the host of <code>-url</code> .
<code>-insecure</code>	—	Skip TLS verification (default <code>true</code> ; SSV ships self-signed certs).
<code>-bases</code>	<code>SSV_BASES</code>	Comma-separated backup IPs seeded before the first scrape.
<code>-backup-cidrs</code>	<code>SSV_BACKUP_CIDRS</code>	CIDR allowlist for discovered backups. Default: primary's <code>/24</code> .
<code>-retries</code>	—	Retries after every endpoint has failed transiently once (default <code>2</code>).
<code>-retry-delay</code>	—	Initial backoff (default <code>200ms</code>); doubles, capped 2 s, $\pm 50\%$ jitter.
<code>-perf-workers</code>	—	Concurrent <code>/performance/{id}</code> calls per scrape (default <code>8</code>).
<code>-listen</code>	—	Exporter HTTP listen address, e.g. <code>:9876</code> .
<code>-ping</code>	—	Probe <code>/serverGroups</code> , print the response, exit.
<code>-install</code> / <code>-uninstall</code>	—	Register / remove the Windows service and exit.
<code>-version</code>	—	Print version and exit.

Precedence when several sources set the same value:

```
explicit flag > env var (flag default) > YAML > built-in default
```

Unknown YAML keys are rejected at load time, so a typo can't silently leave a setting at its default.

Command cheat-sheet

All commands run from `deploy/` .

```
# Bring up / tear down
docker compose up -d                                # Scenario A
docker compose --profile full up -d --build          # Scenario B
docker compose --profile full --profile clip up -d --build # B + C
docker compose down                                # stop, keep data
docker compose down -v                             # stop AND wipe TSDB + Grafana

# Status & logs
docker compose ps
docker compose logs -f prometheus
docker compose logs -f exporter                    # only with --profile full

# Apply an .env change to one service
docker compose up -d --force-recreate prometheus

# Rebuild only the exporter image after a code change
docker compose --profile full build exporter
docker compose --profile full up -d exporter

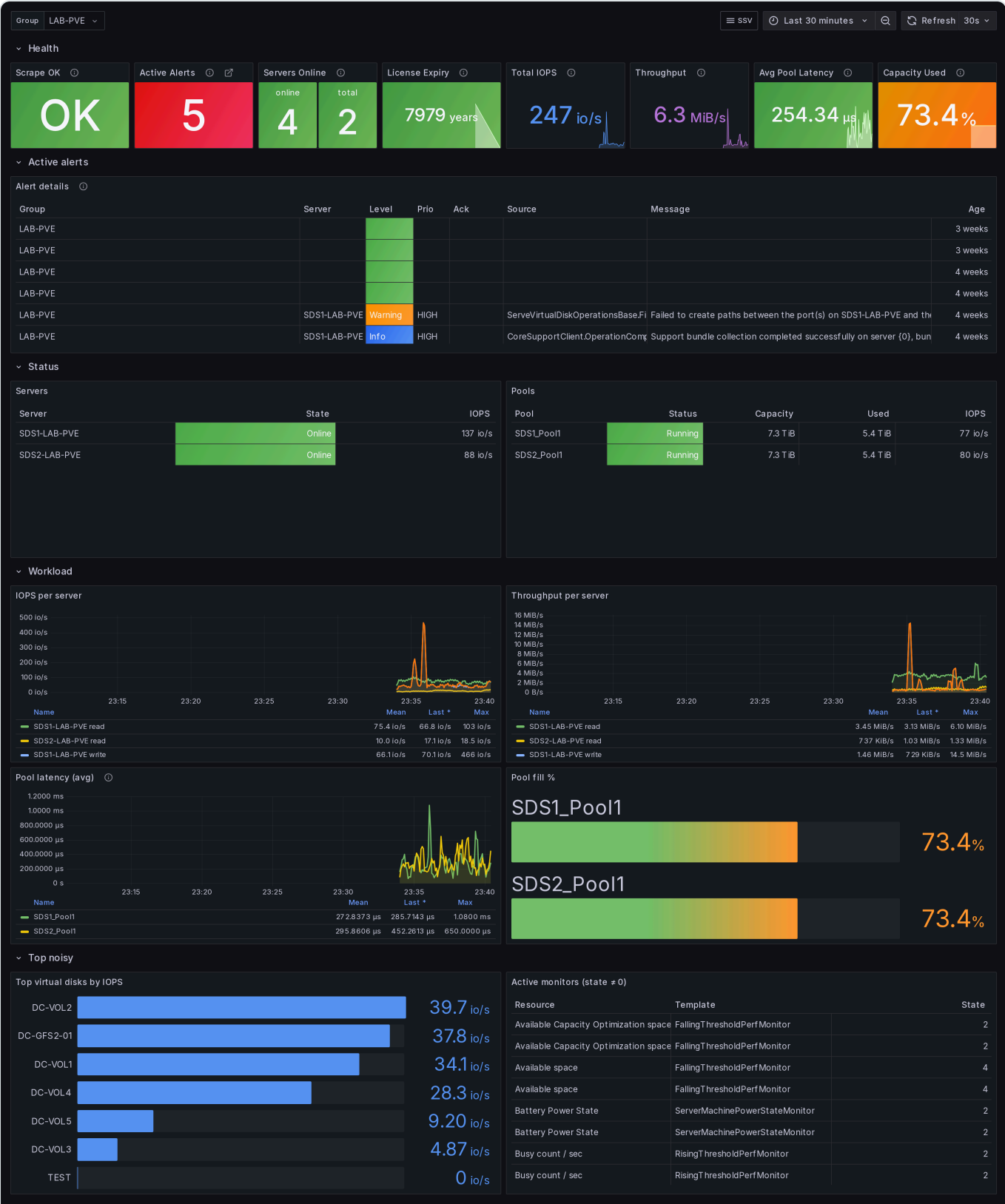
# Reload Prometheus config without a restart (lifecycle API is enabled)
curl -X POST http://localhost:9090/-/reload
```

The Grafana dashboards

Grafana is provisioned with a datasource and **five dashboards** in the **SSV** folder, all cross-linked through an "SSV" dropdown that preserves the time range and selected filters as you move between them. Every panel is filtered by a **Group** template variable, so several SANsymphony groups can sit side by side.

Overview

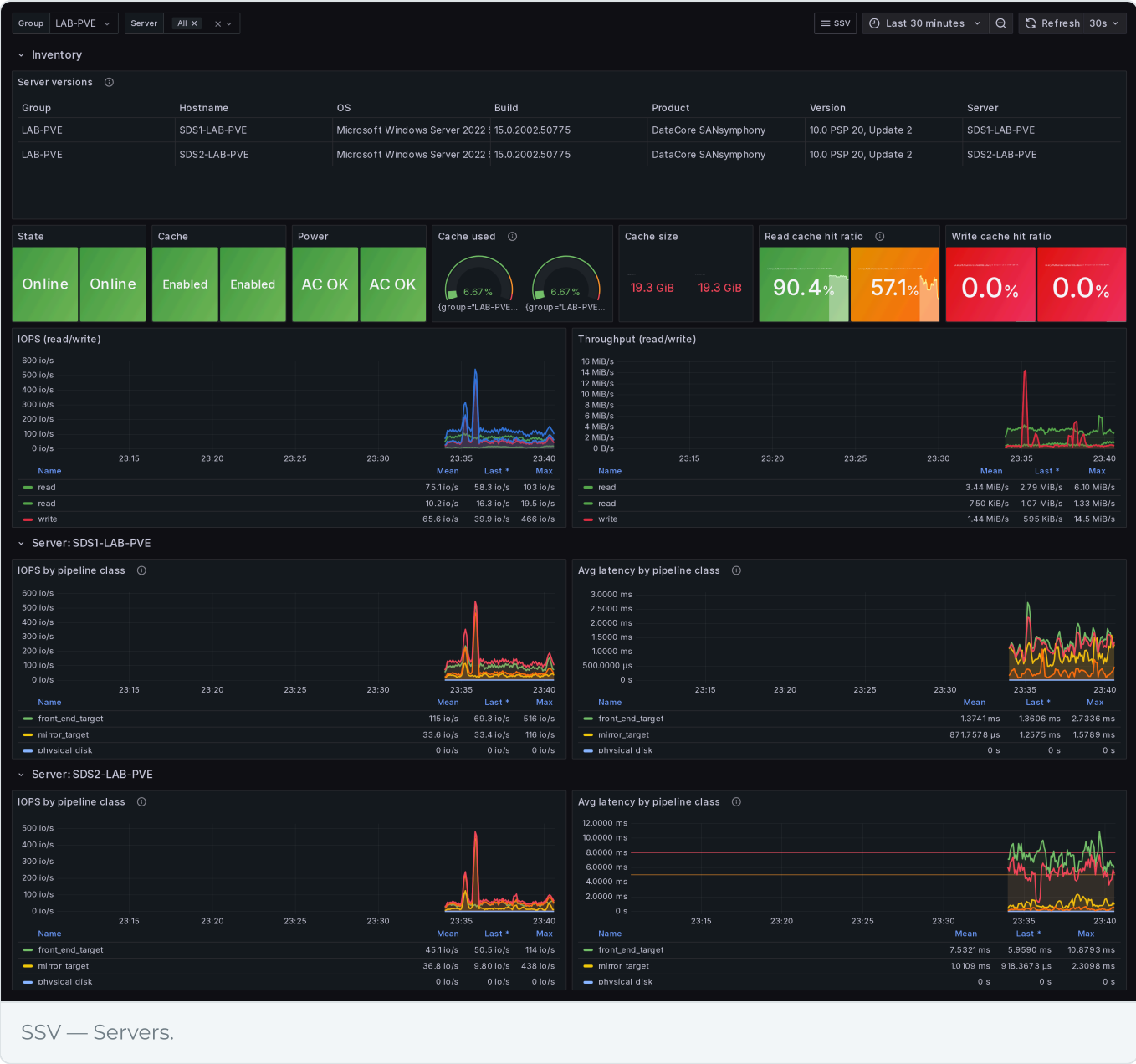
Global health: scrape status, active alerts (level / server / age), server states, capacity rollups, total IOPS & latency, top-N noisy vdisks, active monitors.



SSV — Overview, live PSP 20 lab data.

Servers

Per-server (repeated row): state, cache, IOPS & throughput, and IOPS & latency broken down by IO pipeline class (front-end target / mirror target / back-end / pool / target). A "Server versions" table reads product / OS / build from `ssv_server_info` .



Server: SDS1-LAB-PVE

IOPS by pipeline class

Name	Mean	Last	Max
front_end_target	115 io/s	69.3 io/s	516 io/s
mirror_target	33.6 io/s	33.4 io/s	116 io/s
physical disk	0 io/s	0 io/s	0 io/s

Avg latency by pipeline class

Name	Mean	Last	Max
front_end_target	1.3741 ms	1.3606 ms	2.7336 ms
mirror_target	871.7578 μs	1.2575 ms	1.5789 ms
physical disk	0 s	0 s	0 s

Server: SDS2-LAB-PVE

IOPS by pipeline class

Name	Mean	Last	Max
front_end_target	45.1 io/s	50.5 io/s	114 io/s
mirror_target	36.8 io/s	9.80 io/s	438 io/s
physical disk	0 io/s	0 io/s	0 io/s

Avg latency by pipeline class

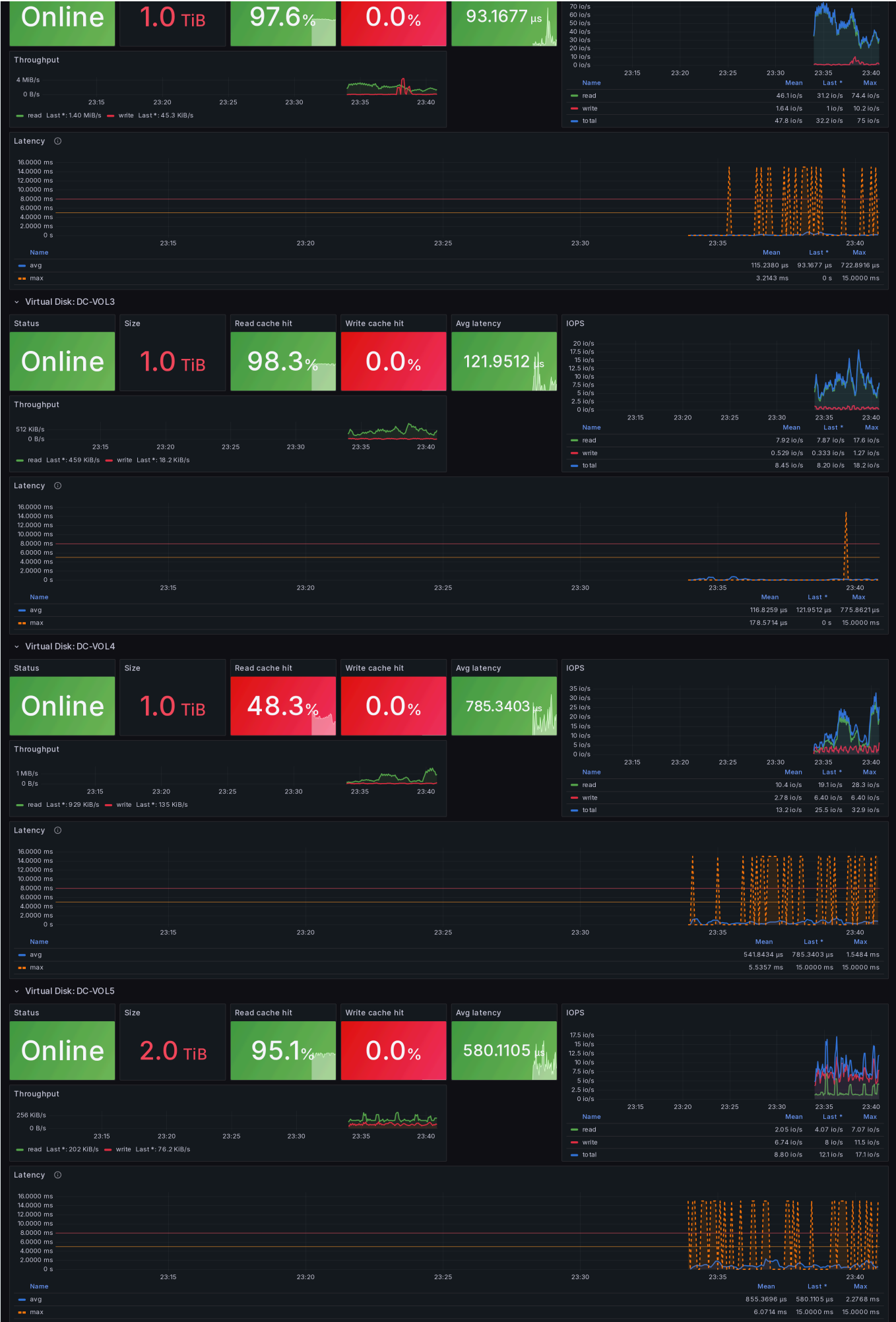
Name	Mean	Last	Max
front_end_target	7.5321 ms	5.9590 ms	10.8793 ms
mirror_target	1.0109 ms	918.3673 μs	2.3098 ms
physical disk	0 s	0 s	0 s

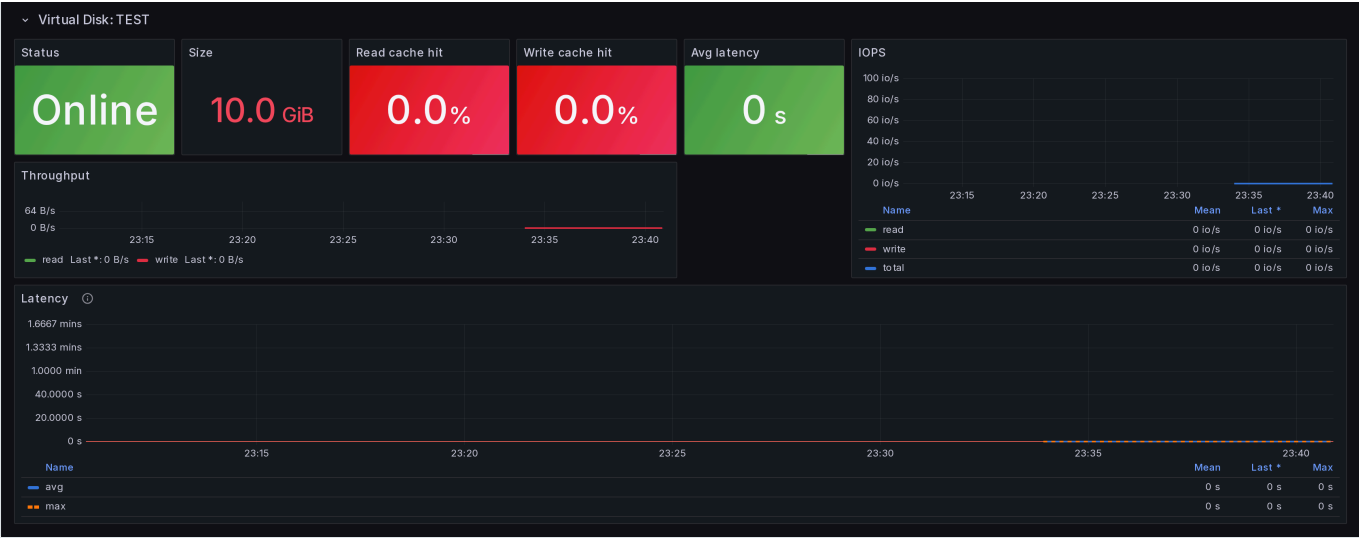
SSV — Servers.

Storage

Per-pool (status, capacity, IOPS, latency) with a collapsible Physical-disks subsection (table + per-disk IOPS / throughput / latency / queue), and per-vdisk (status, cache hit ratio, IOPS, throughput, latency).







SSV — Storage.

Hosts

SAN-client inventory + per-host IOPS & bandwidth, peak IO size, provisioned capacity, plus a Connections subsection for the host's ports.

GroupLAB-PVEHostAll X

SSVLast 30 minutesRefresh 30s

Inventory

Hosts

Group	Host	Hostname	State	Maint	IOPS	B/s
LAB-PVE	pve-lab1	pve-lab1	Online	no	85 io/s	3.6 MiB/s
LAB-PVE	pve-lab2	pve-lab2	Online	no	31 io/s	687.5 KiB/s

Workload

IOPS per host

Name	Mean	Last *	Max
pve-lab1 read	78.4 io/s	69.9 io/s	107 io/s
pve-lab2 read	5.51 io/s	6.13 io/s	7.47 io/s
pve-lab1 write	51.4 io/s	15.2 io/s	445 io/s
pve-lab2 write	25.7 io/s	24.6 io/s	41.3 io/s

Bandwidth per host

Name	Mean	Last *	Max
pve-lab1 read	3.60 MiB/s	3.38 MiB/s	6.18 MiB/s
pve-lab2 read	515 KiB/s	448 KiB/s	691 KiB/s
pve-lab1 write	1.16 MiB/s	268 KiB/s	14.1 MiB/s
pve-lab2 write	277 KiB/s	239 KiB/s	468 KiB/s

IO size

Max read size

Name	Mean	Last *	Max
pve-lab1	152 KiB	128 KiB	1 MiB
pve-lab2	54.3 KiB	64 KiB	128 KiB

Max write size

Name	Mean	Last *	Max
pve-lab1	276 KiB	260 KiB	1 MiB
pve-lab2	137 KiB	152 KiB	648 KiB

Provisioning

Provisioned per host

pve-lab1

10 TiB

pve-lab2

10 TiB

Connections (ports)

Ports of the selected host(s)

Host	Port	Type	Mode	PortName	Conn	IOPS	B/s
SDS1-LAB-PVE	Loopback Port 1	FC	3	20-04-44-9F-7F-98-F8-08	yes	0 io/s	0.0 B/s
SDS2-LAB-PVE	Loopback Port 1	FC	3	20-00-44-9F-7F-A5-45-10	yes	0 io/s	0.0 B/s
SDS1-LAB-PVE	Loopback Port 2	FC	3	20-00-44-9F-7F-98-F8-08	yes	0 io/s	0.0 B/s
SDS2-LAB-PVE	Loopback Port 2	FC	3	20-02-44-9F-7F-A5-45-10	yes	0 io/s	0.0 B/s
SDS1-LAB-PVE	Loopback Port 3	FC	3	20-02-44-9F-7F-98-F8-08	yes	0 io/s	0.0 B/s
SDS2-LAB-PVE	Loopback Port 3	FC	3	20-04-44-9F-7F-A5-45-10	yes	0 io/s	0.0 B/s

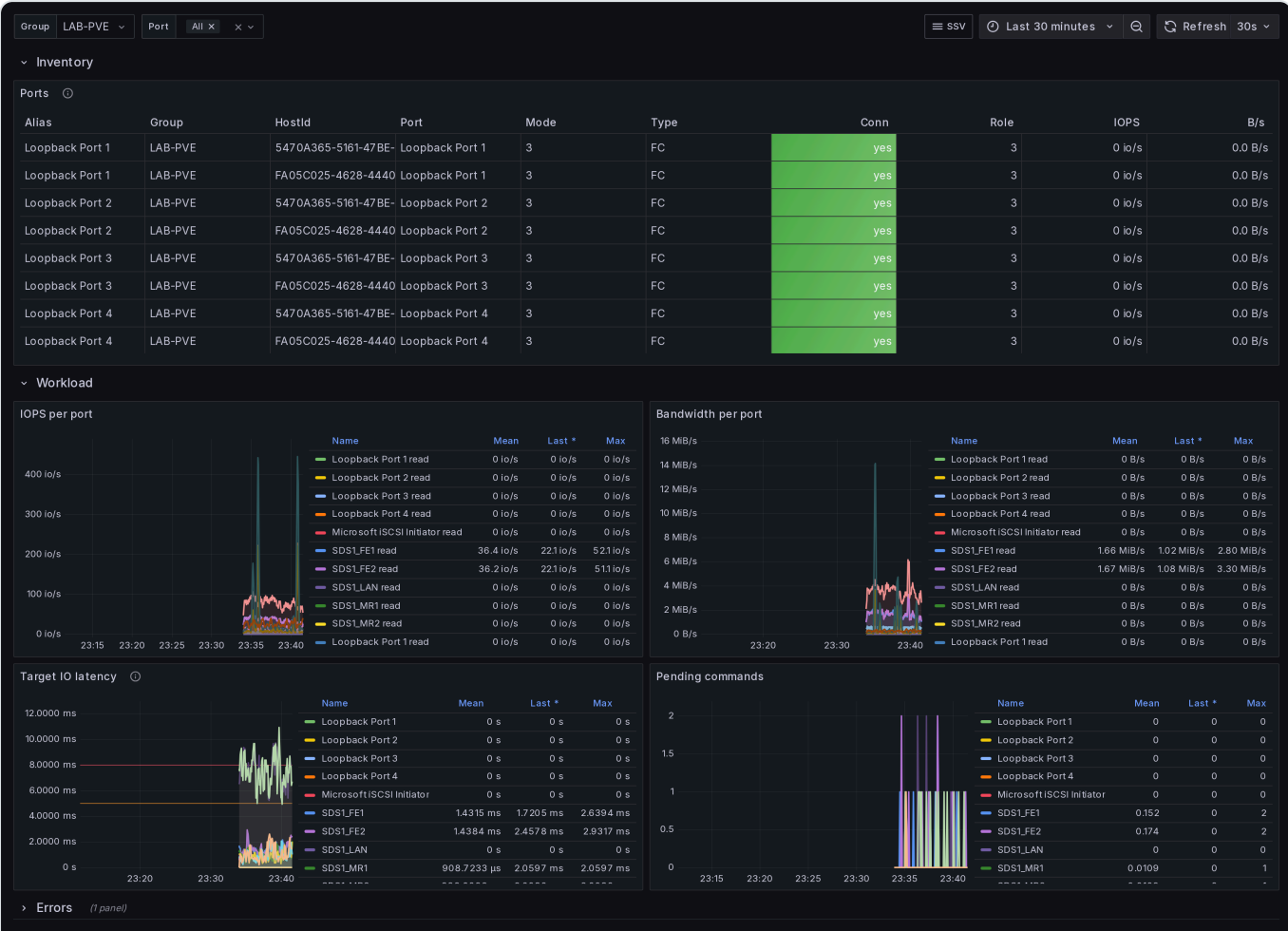
IOPS per port

Name	Mean	Last *
SDS1-LAB-PVE / Loopback Port 1 read	0 io/s	0 io/s
SDS1-LAB-PVE / Loopback Port 2 read	0 io/s	0 io/s
SDS1-LAB-PVE / Loopback Port 3 read	0 io/s	0 io/s
SDS1-LAB-PVE / Loopback Port 4 read	0 io/s	0 io/s
SDS1-LAB-PVE / Microsoft iSCSI Initiator read	0 io/s	0 io/s
SDS1-LAB-PVE / SDS1_FE1 read	36.7 io/s	30.2 io/s
SDS1-LAB-PVE / SDS1_FE2 read	36.6 io/s	30.6 io/s
SDS1-LAB-PVE / SDS1_LAN read	0 io/s	0 io/s
SDS1-LAB-PVE / SDS1_MR1 read	0 io/s	0 io/s
SDS1-LAB-PVE / SDS1_MR2 read	0 io/s	0 io/s
SDS2-LAB-PVE / Loopback Port 1 read	0 io/s	0 io/s

Bandwidth per port

Name	Mean	Last *
SDS1-LAB-PVE / Loopback Port 1 read	0 B/s	0 B/s
SDS1-LAB-PVE / Loopback Port 2 read	0 B/s	0 B/s
SDS1-LAB-PVE / Loopback Port 3 read	0 B/s	0 B/s
SDS1-LAB-PVE / Loopback Port 4 read	0 B/s	0 B/s
SDS1-LAB-PVE / Microsoft iSCSI Initiator read	0 B/s	0 B/s
SDS1-LAB-PVE / SDS1_FE1 read	1.67 MiB/s	1.39 MiB/s
SDS1-LAB-PVE / SDS1_FE2 read	1.68 MiB/s	1.52 MiB/s
SDS1-LAB-PVE / SDS1_LAN read	0 B/s	0 B/s
SDS1-LAB-PVE / SDS1_MR1 read	0 B/s	0 B/s
SDS1-LAB-PVE / SDS1_MR2 read	0 B/s	0 B/s
SDS2-LAB-PVE / Loopback Port 1 read	0 B/s	0 B/s

Per-port (table + IOPS + bandwidth + target IO latency + pending commands) with a collapsible Errors row (link-layer counters).



SSV — Ports.

Editing. Anonymous access is read-only Viewer. Log in as `admin` with `GF_ADMIN_PASSWORD` to edit.

Exposed metrics

Non-exhaustive — see `/metrics` for the live list.

Scrape framing

- `ssv_up{collector="inventory"|"health"|"performance"}` — 1 if the last scrape of that tier succeeded.
- `ssv_scrape_duration_seconds{collector="..."}` — last scrape duration, per tier.

Inventory

`ssv_server_group_*`, `ssv_server_*` (with an `info` series carrying `host_name` / `product_name` / `product_version` / `product_build` / `os_version`), `ssv_pool_*`, `ssv_virtual_disk_*`, `ssv_host_*`, `ssv_port_*`, and `ssv_physical_disk_*` plus a `ssv_physical_disk_pool{disk_id, pool_id, pool, tier}` relation gauge (the `pool` / `tier` labels are also stamped on every disk metric, so PromQL filters need no joins).

Health

`ssv_monitor_state{...}`, `ssv_alerts_total`, `ssv_alert_info{alert_id, machine, level, message, ...}` (gauge=1 per alert), `ssv_alert_age_seconds{alert_id}`.

Performance

Per object: `read_bytes_total`, `write_bytes_total`, `read_ops_total`, `write_ops_total`, plus per-object extras (server cache hits/misses + cache size/free; pool capacity / used / available / reserved / reclamation / oversubscribed; vdisk cache counters; host provisioned + peak IO sizes; port per-direction split + link-layer error counters; physical-disk queue + pending commands).

SSV exposes `*Time` counters in milliseconds; the exporter multiplies by `1e-3` so every latency metric is in seconds. The server tier is split by IO pipeline class:

```
# average IO latency per class
rate(ssv_server_class_io_time_seconds_total[$__rate_interval])
/
rate(ssv_server_class_io_operations_total[$__rate_interval])
```

prom-clip in depth

`prom-clip` ships in this repo as a second binary. It **clips a time window from one Prometheus and replays it into another** — to hand a dataset to support, seed a demo, or carry a before/after comparison between sites that aren't network-connected. Wire format: gzipped OpenMetrics (`.txt.gz`); replay uses the remote-write protocol.

Export

Pick a time window and a step, optionally filter metric names with a regex (e.g. `^ssv_.*`), and write a gzipped OpenMetrics file. Optionally push it to a configured S3 target on completion.

prom-clip

Connection

Export

Import

S3 target

Status

Export a time window to OpenMetrics

Reading from `http://localhost:9090`. [Change.](#)

From (local time)

To (local time)

03/06/2026 22:43

03/06/2026 23:43

Last hour

Last 24h

Last 7d

Last 30d

Step (resolution)

Metric name regex (optional)

30s

ssv_.*

Go duration: 15s, 1m, 5m ...

Anchored on both ends.

Output folder

Browse...

/home/claude/.local/state/prom-clip/exports

Leave empty to use the default `/home/claude/.local/state/prom-clip/exports`.

Output filename

prom-clip-{timestamp}.txt.gz

If you put a path separator in the filename or supply an absolute path, it overrides the folder above. The file can also be downloaded to your browser from the [Status](#) page once the export finishes.

☐ Also upload to the configured [S3 target](#) when the export finishes

Start export

Preview matching metrics

Export — reading from the connected Prometheus, with a `ssv_.*` metric filter and a 30 s step.

Import

Stream a previously exported `.txt.gz` back and replay it into the target Prometheus via remote-write.

prom-clip

ConnectionExportImportS3 targetStatus

Import an OpenMetrics file via remote write

Writing to `http://localhost:9090` at `/api/v1/write` . [Change](#). The Prometheus must have been started with `--web.enable-remote-write-receiver` .

Upload a file from this browser

Choose File

No file chosen

The file is streamed to `/home/claude/.local/state/prom-clip/exports/uploads/` on the server before the import starts.

...or use a file already on the server

prom-clip-20260520-140000.txt.gz

Browse...

Relative names are looked up in `/home/claude/.local/state/prom-clip/exports` . Files produced by Export carry their own timestamps; the target will reject samples older than its out-of-order ingestion window unless `tsdb.out_of_order_time_window` is set high enough.

Batch size (series per WriteRequest)

500

Start import

Import — replays a clip into the target Prometheus over the remote-write protocol.

Status

The last source/target connection and the recent run history.

prom-clip

ConnectionExportImportS3 targetStatus

Runs

No runs yet. Try an [export](#).

Status — run history and the resolved runtime mode (ephemeral vs persistent).

One-shot CLI

The same binary runs headless, with no server, port or state directory:

```
prom-clip export -src http://prom-source:9090 \  
                -from -1h -to now -step 30s \  
                -metric '^ssv_.*' \  
                -out snapshot.txt.gz  
prom-clip import -dst http://prom-target:9090 \  
                -in snapshot.txt.gz
```

`-from / -to` accept RFC3339, a Go duration (`-1h` = "1 hour ago"), or `now` . Pass `-state-dir <path>` to opt into persistent mode (last connection remembered, exports kept, rotation via `-keep-exports N`).

Import requires the target Prometheus to run with `--web.enable-remote-write-receiver` **and** `storage.tsdb.out_of_order_time_window` set wide enough to cover the imported window. The bundled stack exposes both behind `PROM_REMOTE_WRITE` (see Scenario C).

Running the exporter without compose

The compose `exporter` service is convenient, but production deployments usually run the exporter as a long-lived service on each SAN host.

Plain docker run

```
docker run --rm -p 9876:9876 \
  -e SSV_URL=https://10.0.0.1 \
  -e SSV_USER=administrator \
  -e SSV_PASS='ChangeMe!' \
  ghcr.io/lblanc/ssv-prom-exporter:latest
```

Multi-arch images (`linux/amd64` + `linux/arm64`) are published on every release as `vX.Y.Z` , `X.Y` and `latest` . ~34 MB, nonroot uid 65532, `tini` for clean SIGTERM, `HEALTHCHECK` on `/metrics` .

Linux (systemd)

```
tar xzf ssv-prom-exporter-vX.Y.Z-linux-amd64.tar.gz
cd ssv-prom-exporter-vX.Y.Z-linux-amd64
./install-linux.sh
$EDITOR /etc/ssv-prom-exporter/config.yaml # set url / user / pass
systemctl enable --now ssv-prom-exporter
journalctl --unit ssv-prom-exporter -f
```

The unit runs as a `DynamicUser` with `ProtectSystem=strict` , `NoNewPrivileges` , a `SystemCallFilter` and tight memory limits.

Windows (native service / MSI)

From an **elevated** prompt:

```
:: 1. Install the MSI
msiexec /i ssv-prom-exporter-X.Y.Z-x64.msi /qn

:: 2. Place config.yaml in ProgramData and tighten ACLs
copy "C:\Program Files\ssv-prom-exporter\config.example.yaml" ^
"C:\ProgramData\ssv-prom-exporter\config.yaml"
icacls "C:\ProgramData\ssv-prom-exporter\config.yaml" /inheritance:r ^
/grant:r SYSTEM:F Administrators:F

:: 3. Register and start. Only -config lands in the SCM ImagePath.
"C:\Program Files\ssv-prom-exporter\ssv-prom-exporter.exe" ^
-install -config "C:\ProgramData\ssv-prom-exporter\config.yaml"
sc start ssv-prom-exporter
```

The MSI does **not** register the service (so config paths and credentials never leak into MSI properties); `-install` does, baking only the explicitly-set flags into the SCM `ImagePath` .

Service-mode logs go to **Windows Logs** → **Application** under the service's Event Log source.

High availability & multi-group

Failover

The exporter falls over to a backup management server when the primary is unreachable:

- After each successful inventory scrape, every IP from `/servers[].IpAddresses` is added to the backup list (filtered by `SSV_BACKUP_CIDRS`, default = the primary's `/24`).
- On a transient failure (network error, timeout, HTTP 5xx) the next endpoint is tried. HTTP 4xx is **not** a failover trigger — it's a config bug.
- The last-known-good endpoint is sticky for 5 min, so during an outage only the first call pays the dial-timeout cost; after 5 min the next call retries the primary, detecting recovery.
- The `ServerHost` header is rewritten per endpoint (each backup uses its own IP); SSV rejects hostname-based `ServerHost` values with HTTP 400.

Seed the backup list before the first scrape with `SSV_BASES=ip1,ip2, ...`.

Multi-group

SSV management servers federate their state: `/serverGroups` returns the local group plus every linked peer, and `/servers` mixes local nodes with remote ones, but `/performance/{id}` is local-only. The exporter therefore scrapes per-server inventory and performance **only for the local group** (`OurGroup=true`). **Run one exporter per SSV group**, and list them all in `EXPORTER_TARGETS`:

```
EXPORTER_TARGETS=HCI104=10.12.104.121:9876,HCI130=10.12.130.121:9876
```

Each becomes a Prometheus target carrying its `group=<name>` label, and the dashboards filter on `$group` end-to-end.

Troubleshooting & gotchas

Symptom	Cause / fix
ssv_up=0 on all collectors, fast-fail	Bad credentials, wrong <code>ServerHost</code> , or a stale session token. Check <code>docker compose logs exporter</code> .
HTTP 400 / <code>ErrorCode 9</code>	The mandatory <code>ServerHost</code> header is missing or set to a hostname. It must be the IP being hit — hostnames are rejected even when they resolve.
Grafana panels empty for a group	The Group dropdown doesn't match a scraped target, or that exporter's <code>performance</code> tier is down. Check Prometheus targets.
prom-clip import "succeeds" but nothing lands	Target Prometheus has no out-of-order window — set <code>PROM_REMOTE_WRITE=1</code> + <code>PROM_OOO_WINDOW</code> .
Data looks stale	SSV's REST cache is 30 s by default (<code>RequestExpirationTime</code> in the mgmt server's <code>Web.config</code>); scraping faster sees stale data.

Releases

Releases are produced by GitHub Actions on an annotated `v*` tag. Each release ships the Windows `.exe`, the MSI, a Linux tarball, `SHA256SUMS`, and multi-arch GHCR images for both `ssv-prom-exporter` and `prom-clip`. The latest published release is **v0.8.1**; see the project's GitHub Releases for the full list and the `CHANGELOG.md` for per-version notes.

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